

Homework 3

In this homework, we learn how to use generative adversarial network (GAN) to generate painting images. You are encouraged to implement GAN from scratch by yourself, to learn how to implement a deep generative model. However, if you already know how to implement GAN or would like to save some time to focus more on your course project, it is also OK to use an existing implementation from GitHub.

Then, you can take the painting image dataset from <https://www.kaggle.com/datasets/steubk/wikiart> and train a GAN model on it. Use the trained GAN to generate painting images, and write a report (2 pages, NeurIPS format). In the report, you can include the following: 1) hyperparameter settings; 2) training loss curves; 3) Inception score and FID score of generated images, using code from <https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/> 4) examples of generated images. Please submit the report to Gradescope.